

FAI Capstone: Final Report

A Reproducibility Study of A* Search: Heuristic Sensitivity and Practical Performance Tradeoffs

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Recorded Demonstration Link: Insert YouTube/Vimeo Link Here

Code Repository Link: Insert GitHub Repository Link Here

Introduction

Pathfinding and graph search are foundational problems in Artificial Intelligence. Many planning, robotics, logistics, routing, and game systems depend on efficiently discovering low-cost paths through structured environments. A central challenge in search is balancing two competing objectives: solution optimality and computational efficiency. Algorithms that guarantee optimality often require substantial exploration, while faster methods may sacrifice correctness.

The landmark paper by Hart, Nilsson, and Raphael (1968), *A Formal Basis for the Heuristic Determination of Minimum Cost Paths*, introduced the A* algorithm as a principled solution to this tradeoff. By combining exact path cost with heuristic estimates of remaining distance, A* was designed to preserve optimality while reducing unnecessary exploration.

This paper was selected because it is both historically significant and deeply connected to core course themes involving search, admissibility, heuristic reasoning, and rational decision-making. It also provides an ideal setting for reproducibility analysis because its theoretical claims are clear and testable in modern computational environments.

This project reproduces and evaluates the central claims of the original A* paper through controlled Gridworld experiments, while extending the analysis to study heuristic sensitivity through Weighted A* variants.

Paper Overview

The original A* paper proposed a general best-first search framework using the evaluation function:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ is the true cost from the start state to node n

- $h(n)$ is an estimate of the remaining cost from n to the goal

The key insight was that search should be guided by both accumulated cost and informed prediction rather than either quantity alone.

The major claims of the original paper include:

- If $h(n)$ is admissible (never overestimates), A* returns an optimal solution.
- Better heuristics reduce unnecessary exploration.
- A* is optimally efficient among optimal algorithms using the same heuristic information.

These claims established A* as one of the most influential search algorithms in AI.

Reproduction Objective

The goal of this project was not to recreate the original 1968 hardware or exact experimental environment, but to reproduce the algorithmic claims in a modern controlled setting.

The specific reproduction objectives were:

1. Verify that standard A* with an admissible heuristic matches the optimal path cost of Uniform Cost Search (UCS).
2. Measure whether A* reduces search effort relative to UCS.
3. Study how increasing heuristic aggressiveness affects performance through Weighted A*.
4. Evaluate behavior across environments of increasing difficulty.

Thus, the project reproduces the theoretical behavior of A* while extending the study to practical heuristic tradeoffs.

Methodology

A full experimental framework was implemented in Python.

The environment consists of procedurally generated 2D Gridworlds with:

- four-direction movement (up, down, left, right)
- unit step cost
- random blocked cells representing obstacles
- fixed start at $(0, 0)$ and goal at bottom-right corner

The following algorithms were implemented:

- **Uniform Cost Search (UCS):** equivalent to A* with $h(n) = 0$
- **Standard A*:** Manhattan heuristic

- **Weighted A*:**

$$f(n) = g(n) + \epsilon h(n), \quad \epsilon > 1$$

Implementation details included:

- priority queue frontier using binary heaps
- closed-set duplicate detection
- parent-pointer path reconstruction
- reproducible random seeds

The Manhattan heuristic was selected because it is admissible and consistent for four-direction Gridworld movement.

Experiments

Three environment tiers were evaluated:

Grid Size	Obstacle Density	Difficulty Level
15 × 15	0.10	Easy
20 × 20	0.20	Medium
30 × 30	0.25	Hard

Each configuration used 10 independent random seeds.

Weighted A* was tested with:

$$\epsilon \in \{1.5, 2.0, 3.0\}$$

The following metrics were recorded:

- Path cost
- Nodes expanded
- Runtime (ms)
- Peak frontier size
- Solved instances

Results

Easy Environment (15x15, Density 0.10)

Algorithm	Solved	AvgCost	AvgExpanded	Runtime	PeakFrontier
UCS	10	28.00	199.60	1.2701	16.20
A*	10	28.00	173.40	1.2034	25.20
WA* 1.5	10	28.60	32.00	0.3229	29.40
WA* 2.0	10	28.80	31.60	0.3076	29.10
WA* 3.0	10	29.20	31.00	0.2371	28.70

Medium Environment (20x20, Density 0.20)

Algorithm	Solved	AvgCost	AvgExpanded	Runtime	PeakFrontier
UCS	9	38.67	281.90	1.5444	17.50
A*	9	38.67	187.60	1.2460	29.30
WA* 1.5	9	39.56	45.30	0.2990	33.20
WA* 2.0	9	40.22	43.00	0.2772	32.80
WA* 3.0	9	40.67	42.40	0.2557	31.90

Hard Environment (30x30, Density 0.25)

Algorithm	Solved	AvgCost	AvgExpanded	Runtime	PeakFrontier
UCS	4	59.50	460.60	2.8340	21.60
A*	4	59.50	334.60	2.2911	41.80
WA* 1.5	4	61.00	232.80	1.7570	42.70
WA* 2.0	4	62.50	232.20	1.7958	38.30
WA* 3.0	4	62.50	231.40	1.7741	37.90

Analysis and Discussion

Several strong trends emerge.

First, across all solved environments, standard A* matched the path cost of UCS exactly. This strongly supports the classical claim that admissible heuristics preserve optimality.

Second, A* consistently reduced node expansions relative to UCS:

- Easy: 199.60 $\rightarrow$ 173.40
- Medium: 281.90 $\rightarrow$ 187.60
- Hard: 460.60 $\rightarrow$ 334.60

This confirms that heuristic guidance reduces unnecessary exploration.

Third, Weighted A* produced dramatic efficiency gains. In medium difficulty environments, expansions dropped from 281.90 (UCS) to approximately 42–45. Runtime was also substantially reduced.

However, these gains came with higher path costs, and the degradation became more visible in harder environments. This demonstrates a practical tradeoff:

Higher heuristic weight $\Rightarrow$ Faster search but weaker optimality

Fourth, solvability decreased as difficulty increased. Only 4/10 hard environments were solvable, showing that obstacle density strongly influences practical planning difficulty.

An interesting secondary result is that peak frontier size did not always decrease alongside expansions. This suggests that memory behavior and search effort are related but distinct metrics.

Challenges in Reproduction

Several practical challenges were encountered:

- The original paper does not specify modern benchmark environments comparable to Gridworld.
- Exact historical hardware/runtime comparisons are impossible.
- Random obstacle generation can create unsolvable maps.
- Memory behavior required separate measurement beyond runtime alone.

These issues are common in reproducing foundational research papers.

Limitations

This study has several limitations:

- Gridworld is simpler than many real planning domains.
- Only Manhattan heuristics were evaluated.
- Runtime values depend on implementation language and hardware.
- Weighted A* variants are an extension beyond the original paper.

Nevertheless, these simplifications preserve the central logic of heuristic search.

Reproducibility Verdict (Required)

Verdict: Partially Reproducible

The central theoretical claims of the paper were successfully reproduced:

- A* preserved optimality relative to UCS.
- A* reduced search effort.
- Heuristic quality strongly affected efficiency.

However, the exact historical experimental setup, datasets, and computational context of the 1968 paper were not fully recoverable. Therefore, the paper is best classified as partially reproducible rather than fully reproducible.

Conclusion and Future Work

This project successfully reproduced the core algorithmic behavior of A* search in a modern experimental framework. Standard A* consistently achieved the same solution quality as UCS while requiring less exploration and lower runtime. Weighted A* further improved efficiency at the cost of optimality.

The study reinforces the enduring importance of heuristic design in Artificial Intelligence. More than fifty years after publication, the central insight of the original A* paper remains highly relevant: informed search can dramatically reduce computation while preserving correctness when heuristics are carefully designed.

Future work could include alternative heuristics, dynamic environments, larger map sets, diagonal movement, and real-world routing benchmarks.

References

- Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2), 100–107.
- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.